Open Access

Supplementary Information — NMDA Modulates Working Memory Attractor Stability

Opposite Regimes Can Produce Schizophrenia-like Instability and OCD-like Overstability

H. Tahiri, **R. Zare**, **F. Amidizade**, **I. Chakir**, and **A. Kumar**

Abstract

This document contains the supplementary figures, tables, and extended methods supporting the main manuscript regarding NMDA modulation of working memory attractors.

Keywords working memory, attractor networks, NMDA, schizophrenia, OCD

1. Supplementary Figures

This supplement gives the complete mathematical description of the spiking network (Section 1), the exact parameter values used in all simulations (Table 1–Table 4), the definition of the λ_{delay} metric (Section 2), the multi-seed statistical protocol (Section 3), a supplementary NMDA-conductance sweep (Figure 1), and the full statistical tables (Section 4). The model follows the integrate-and-fire framework of (Loh et al., 2007) as used by (Loh et al., 2007),and was implemented in Brian2 (Stimberg et al., 2019).

1.a. S1. Network model

1.a.i. *S1.1 Single-neuron dynamics:*

Both excitatory (E) and inhibitory (I) cells are leaky integrate-and-fire neurons. The subthreshold membrane potential V obeys

$$C_m \frac{dV}{dt} = -g_L (V - E_L) - I_{\text{syn}}(t), \tag{1}$$

with cell-type-specific C_m and g_L (Table 1). When V reaches the threshold V_{th} a spike is emitted and V is clamped to the reset V_r for an absolute refractory period t_{ref}. The total synaptic current is

$$I_{\text{syn}}(t) = I_{\text{AMPA,ext}}(t) + I_{\text{AMPA,rec}}(t) + I_{\text{NMDA,rec}}(t) + I_{\text{GABA}}(t). \tag{2}$$

1.a.ii. *S1.2 Synaptic currents:*

AMPA (external and recurrent) and GABA_A currents are

Published May 30, 2026

$$I_{\mathrm{AMPA}}(t) = s_{\mathrm{AMPA}}(t)\left(V(t) - V_E\right),\ \frac{ds_{\mathrm{AMPA}}}{dt} = -\frac{s_{\mathrm{AMPA}}}{\tau_{\mathrm{AMPA}}} + \sum_k g_{\mathrm{AMPA}}\,\delta(t - t_k),$$
$$I_{\mathrm{GABA}}(t) = s_{\mathrm{GABA}}(t)\left(V(t) - V_I\right),\ \frac{ds_{\mathrm{GABA}}}{dt} = -\frac{s_{\mathrm{GABA}}}{\tau_{\mathrm{GABA}}} + \sum_k g_{\mathrm{GABA}}\,\delta(t - t_k), \tag{3}$$

where each presynaptic spike at t_k increments the conductance variable by the corresponding synaptic weight. The NMDA current carries a voltage-dependent Mg $^{2+}$ block (Jahr & Stevens, 1990):

$$I_{\mathrm{NMDA}}(t) = \frac{g_{\mathrm{NMDA}}\,s^{\mathrm{tot}}_{\mathrm{NMDA}}(t)\left(V(t) - V_E\right)}{1 + \frac{[\mathrm{Mg}^{2+}]}{3.57}\exp\left(-0.062\,V(t)\right)}, \tag{4}$$

with V in mV and $[\mathrm{Mg}^{2+}]$ in mM. The NMDA gating variable has a two-stage (rise/decay) form,

$$\frac{ds_{\mathrm{NMDA}}}{dt} = -\frac{s_{\mathrm{NMDA}}}{\tau_{\mathrm{NMDA,decay}}} + \alpha\,x(t)\left(1 - s_{\mathrm{NMDA}}\right),\ \frac{dx}{dt} = -\frac{x}{\tau_{\mathrm{NMDA,rise}}} + \sum_k \delta(t - t_k), \tag{5}$$

and $s^{\mathrm{tot}}_{\mathrm{NMDA}}$ denotes the gating summed over the presynaptic excitatory pool. All synapses act with a transmission delay of 0.5 ms.

$$I_{\mathrm{NMDA}}(t) = \frac{g_{\mathrm{NMDA}} s^{\mathrm{tot}}_{\mathrm{NMDA}}(t)(V(t) - V_E)}{1 + \frac{[\mathrm{Mg}^{2+}]}{3.57}\exp(-0.062V(t))}, \tag{6}$$

with V in mV and $[\mathrm{Mg}^{2+}]$ in mM. The NMDA gating variable has a two-stage (rise/decay) form,

$$\begin{aligned}\frac{ds_{\mathrm{NMDA}}}{dt} &= -\frac{s_{\mathrm{NMDA}}}{\tau_{\mathrm{NMDA,decay}}} + \alpha x(t)(1 - s_{\mathrm{NMDA}}),\\ \frac{dx}{dt} &= -\frac{x}{\tau_{\mathrm{NMDA,rise}}} + \sum_k \delta(t - t_k),\end{aligned} \tag{7}$$

and $s^{\mathrm{tot}}_{\mathrm{NMDA}}$ denotes the gating summed over the presynaptic excitatory pool. All synapses act with a transmission delay of 0.5 ms.

1.a.iii. *S1.3 Architecture and connectivity:*

The network contains $N = 2000$ neurons: $N_E = 1600$ excitatory ($f_{\mathrm{inh}} = 0.20$) and $N_I = 400$ inhibitory. The excitatory population is split into two selective pools S_1 and S_2 of 240 neurons each ($f_{\mathrm{sel}} = 0.15$ of N_E) and a non-selective pool (NS) of 1120 neurons. The inhibitory pool provides global feedback inhibition. Recurrent excitation is structured by a within-pool potentiation factor J_p and a compensating cross-pool factor

$$J_m = 1 - f_{\mathrm{sel}}\,\frac{J_p - 1}{1 - f_{\mathrm{sel}}}, \tag{8}$$

which keeps the mean recurrent input constant as J_p varies. Excitatory–excitatory AMPA weights are g_{EEA} (baseline), $g_{\mathrm{EEA}}\,J_p$ (within a selective pool) and $g_{\mathrm{EEA}}\,J_m$ (between selective pools); recurrent NMDA is pooled analogously. $E \to I$, $I \to E$

and $I \to I$ connections are all-to-all with weights g_{EIA}, g_{IE} and g_{II} respectively. Base conductances (Table 3) are calibrated for $N = 2000$ and rescaled by $1600/N_E$ (excitatory) and $400/N_I$ (inhibitory) for other network sizes.

1.a.iv. *S1.4 Background input and stimulation protocol:*

Every neuron receives $N_{\mathrm{ext}} = 800$ independent external AMPA synapses driven by Poisson spike trains at 3 Hz each (2.4 kHz aggregate), reproducing cortical spontaneous activity. A 200 ms spontaneous period is followed by a cue of 200 Hz applied to S_1 over $t \in [200, 700]$ ms, then a delay period $[700, 1200]$ ms. An optional distractor is applied to S_2 over $[800, 1100]$ ms at either 0 or 200 Hz. Total simulation time is 1200 ms. Equations were integrated with the forward Euler method at $dt = 0.02$ ms; independent trials differ only in random seed.

1.a.v. *S1.5 Modelled regimes:*

NMDA and GABA conductances are scaled multiplicatively relative to control. The three main-text regimes hold g_{GABA} fixed and vary g_{NMDA}: control (1.00), SCZ-like (0.95, -5%) and OCD-like (1.10, $+10\%$). The effective recurrent NMDA conductance is therefore $g_{\mathrm{EEN}} \times \{0.95, 1.00, 1.10\} = \{0.157, 0.165, 0.182\}$ nS.

Parameter	Excitatory	Inhibitory
Resting potential E_L	-70 mV	-70 mV
Threshold V_{th}	-50 mV	-50 mV
Reset V_r	-55 mV	-55 mV
Membrane capacitance C_m	0.5 nF	0.2 nF
Leak conductance g_L	25 nS	20 nS
Membrane time constant $\tau_m = C_m/g_L$	20 ms	10 ms
Refractory period t_{ref}	2 ms	1 ms

Table 1: Single-neuron parameters (excitatory / inhibitory).

Parameter	Value	Parameter	Value
τ_{AMPA}	2 ms	V_E	0 mV
$\tau_{\mathrm{NMDA,rise}}$	2 ms	V_I	-70 mV
$\tau_{\mathrm{NMDA,decay}}$	100 ms	$[\mathrm{Mg}^{2+}]$	1 mM
τ_{GABA}	10 ms	Synaptic delay	0.5 ms
α (NMDA)	0.5 ms^{-1}	Integration step dt	0.02 ms

Table 2: Synaptic kinetics and reversal potentials.

Conductance	Value	Conductance	Value
$g_{\mathrm{ext},E}$ (external AMPA, E)	2.10 nS	g_{EEN} (rec. NMDA, $E \to E$)	0.165 nS
$g_{\mathrm{ext},I}$ (external AMPA, I)	1.62 nS	g_{EIN} (rec. NMDA, $E \to I$)	0.130 nS

g_{EEA} (rec. AMPA, $E \to E$)	0.050 nS	g_{IE} (GABA, $I \to E$)	1.30 nS
g_{EIA} (rec. AMPA, $E \to I$)	0.040 nS	g_{II} (GABA, $I \to I$)	1.00 nS

Table 3: Base synaptic conductances (calibrated for $N = 2000$; the excitatory/inhibitory scale factors equal 1 at this size).

Quantity	Value	Quantity	Value
Total neurons N	2000	Cue (to S_1)	200 Hz, $[200, 700]$ ms
Excitatory N_E / Inhibitory N_I	1600/400	Delay period	$[700, 1200]$ ms
Selective pools S1 = S2	240 each	Distractor (to S_2)	0 or 200 Hz, $[800, 1100]$ ms
Non-selective pool NS	1120	Total runtime	1200 ms
External synapses N_{ext}	800 at 3 Hz	Potentiation J_p	1.84 (swept 1.75–1.90)
Cross-pool factor J_m at $J_p = 1.84$	0.852		

Table 4: Network architecture and stimulation protocol.

1.b. *S2. The λ_{delay} metric*

Population firing rates were obtained from spike-count histograms smoothed with a flat (rectangular) sliding window of width 50 ms. During the delay we form the differential rate $\Delta r(t) = r_{S_1}(t) - r_{S_2}(t)$ and fit a single exponential $\Delta r(t) = A\,e^{-\lambda t}$ to its post-peak segment by linear regression of $\log \Delta r$. The fit is evaluated over the window $[\,t_{\mathrm{off}} + 100\ \text{ms},\ t_{\mathrm{end}} - 50\ \text{ms}\,]$, starting from the peak of Δr within that window so that the post-cue rising transient does not bias the estimate. Small λ_{delay} indicates stable maintenance; large λ_{delay} indicates rapid collapse. We define a persistence zone as $\lambda_{\mathrm{delay}} < 5\ \text{s}^{-1}$ (half-life ≈ 139 ms); equivalently $1/\lambda_{\mathrm{delay}} > 0.2$ s. Networks that never encode the cue or that enter a runaway state return an undefined (NaN) value and are treated as a separate failure category rather than as $\lambda = 0$.

1.c. *S3. Multi-seed statistical protocol*

To characterise stability beyond single realisations, each parameter combination was simulated with 19 independent random seeds. The analysis grid crossed six NMDA/GABA conductance settings — $(g_{\mathrm{NMDA}}/g_{\mathrm{GABA}}) \in \{(0.8/0.4), (0.9/0.3), (1.0/0.3), (1.0/0.6), (1.1/0.4), (1.1/0.6)\}$ — with two recurrent weights ($J_p = 1.75$ and 1.88), run separately with and without the S_2 distractor. Because per-condition λ_{delay} distributions departed from normality (Shapiro–Wilk), non-parametric tests were used throughout: Kruskal–Wallis across conditions (with η^2 effect size), pairwise Mann–Whitney U with Bonferroni correction, and Spearman rank correlations between λ_{delay} and other delay-period metrics. Table 5–Table 8 report the with-distractor analysis used for the main-text statistics.

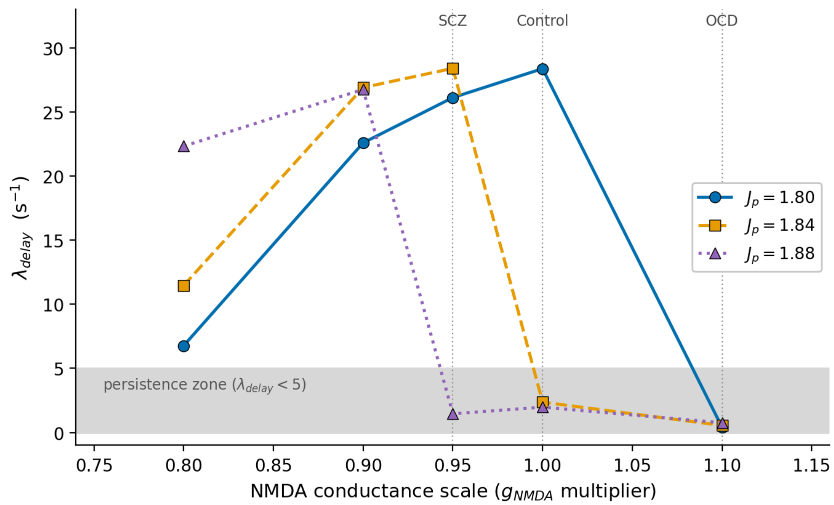

Figure 1: **NMDA-conductance sweep at fixed GABA ($g_{\mathbf{GABA}} = 1.0$).** λ_{delay} as a function of the NMDA conductance multiplier for three recurrent weights J_p. The shaded band is the persistence zone ($\lambda_{\text{delay}} < 5$); dotted reference lines mark the three main-text regimes (SCZ 0.95, control 1.00, OCD 1.10). Increasing NMDA drives the network from non-encoding/collapse at low values, through a high-λ unstable band, into deep persistence at $g_{\text{NMDA}} \geq 1.0$; the transition shifts to lower NMDA as J_p increases. Values below $g_{\text{NMDA}} \approx 0.7$ are omitted because the network fails to encode the cue and λ_{delay} is undefined. Generated from the single-seed grid sweep (`Grid_results_200Hz.csv`).

1.d. S4. Statistical tables

J_p	NMDA	GABA	n	Mean	SD	Median	[Min, Max]
1.75	1.00	0.60	11	32.54	1.82	32.20	[30.28, 37.22]
1.75	1.10	0.40	19	33.99	3.46	34.15	[27.65, 40.83]
1.75	1.10	0.60	19	13.64	11.85	7.23	[2.26, 34.14]
1.88	0.80	0.40	3	33.90	0.61	34.15	[33.21, 34.34]
1.88	0.90	0.30	19	34.50	2.49	34.36	[29.53, 41.16]
1.88	1.00	0.30	19	4.60	8.06	1.65	[0.60, 28.12]
1.88	1.00	0.60	19	0.92	1.13	0.61	[−0.08, 4.97]
1.88	1.10	0.40	19	0.31	0.39	0.27	[−0.33, 1.21]
1.88	1.10	0.60	19	0.26	0.34	0.24	[−0.25, 1.16]

Table 5: Descriptive statistics of λ_{delay} (s^{-1}) in the persistent regime, seed-level means, with distractor. n is the number of seeds (of 19) that reached the persistent regime for that cell.

Grouping	H (df)	p	η^2
$J_p = 1.75$	25.97 (2)	2.30×10^{-6}	0.521
$J_p = 1.88$	75.87 (5)	6.14×10^{-15}	0.770

Combined	40.88 (5)	9.92×10^{-8}	0.254

Table 6: Omnibus Kruskal–Wallis tests on λ_{delay} across conditions (with distractor).

Regime	n	Median λ_{delay}	Mann–Whitney
Persistent	155	3.33	$U = 4515,\ p = 5.6 \times 10^{-5}$
Transient	85	29.21	

Table 7: Overall persistent vs. transient comparison (with distractor): median λ_{delay} and Mann–Whitney U.

Metric	ρ	p
Attractor robustness (min Δr)	−0.952	$< 10^{-80}$
Signal-to-noise ratio (delay)	−0.947	$< 10^{-76}$
Firing-rate variability σ	+0.921	$< 10^{-63}$
Minimum S_1 rate (delay)	−0.910	$< 10^{-59}$
Mean S_1 rate (delay)	−0.838	$< 10^{-41}$
Delay gain	−0.613	$< 10^{-16}$

Table 8: Spearman rank correlations between λ_{delay} and delay-period metrics (persistent regime, with distractor). Negative values indicate that faster decay accompanies weaker maintenance.

Note

λ_{delay}, attractor robustness (min Δr) and several of the metrics in Table 8 are derived from the same $\Delta r(t)$ trajectory; the correlations therefore quantify internal consistency of the metric rather than independent validation. All values were produced by the accompanying code (`Working_Memory_Model.py`, `Run_Grid_Trials_points.py`, `statistical_analysis.py`).

References

Jahr, C. E., & Stevens, C. F. (1990). Voltage dependence of NMDA-activated macroscopic conductances predicted by single-channel kinetics. *Journal of Neuroscience*, *10*(9), 3178–3182.

Loh, M., Rolls, E. T., & Deco, G. (2007). A Dynamical Systems Hypothesis of Schizophrenia. *PLOS Computational Biology*, *3*(11), e228. https://doi.org/10.1371/journal.pcbi.0030228

Stimberg, M., Brette, R., & Goodman, D. F. M. (2019). Brian 2, an Intuitive and Efficient Neural Simulator. *Elife*, *8*, e47314. https://doi.org/10.7554/eLife.47314